# MUHAMMAD LABIYB AFAKH

Tokyo, Japan

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## SUMMARY

Robotics Engineer with 7+ years of experience developing autonomous systems using C++, Python, and C across manipulator, mobile, aerial, , surface vehicle, and humanoid platforms. Combines end-to-end robot development with experience in perception, planning, and control, with recent work focused on learning-based control and sim-to-sim-to-real transfer.

- End-to-end development spanning mechanical design, electrical hardware, embedded software, and system integration to build complete robotic platforms.
- Development and evaluation of autonomous navigation and manipulation algorithms, including collision avoidance, path tracking, object detection, and grasp pose estimation.
- Integration of microcontroller firmware, actuator interfaces, and sensor drivers with high-level software in both ROS/ROS2-based and standalone systems, using shared memory and socket-based communication. Experience in concurrent data acquisition, communication optimization, and debugging across hardware and software.

## PROFESSIONAL EXPERIENCES

**Computer Vision and Robot Engineer** June 2026-now
Kawada Robotics

- Evaluated existing and developed new feature for computer vision system.
- Working on Humanoid Robot.

**Computer Vision and Robot Motion** October 2025-now
FingerVision,Inc. (Intern)

- Built full UR3 robotic evaluation platform from scratch, integrating FingerVision tactile sensors with ROS-based MoveIt motion control and designing experiments to evaluate perception capabilities.
- Evaluated and improved object detection models (YOLO, Mask R-CNN) for grasp object classification and placement pose estimation through systematic benchmarking, collecting and annotating new datasets for fine-tuning to address performance gaps.
- Identified and resolved issues across the existing imitation learning pipeline using Dobot manipulator, improving data collection, training, and inference processes for in-house and client proof-of-concept development.
- Evaluated promising project directions and contributed to technical strategy.

**Robotics Engineer** April 2024 - March 2025
QibiTech Inc. (Part-Time)

- Developed ROS2 driver and MoveIt integration package for HR4C manipulator from scratch, implementing hardware interfaces for position, velocity, and effort control modes with action servers and automated test suite.
- Created robot description (URDF), visualization, and simulation environments in PyBullet and Gazebo for development and validation.
- Debugged and improved system integration between Kachaka AMR and HATS, resolving issues for more reliable autonomous operation.
- Analyzed mobile manipulator manipulability in Gazebo simulation for client project, and evaluated navigation performance using Robotics Toolbox Python for proof-of-concept development.

**Research and Development Engineer** Jan 2022 - March 2024
Shanghai Micro Ears Ingeniousness Intelligent Technology Co., Ltd. (Part-Time)

- Designed and manufactured the robot structure.
- Developed embedded control software and sensor data processing pipeline.
- Optimized serial communication protocol and data packet structure, achieving 200Hz sampling rate with real-time GUI visualization.
- Developed and integrated system with a manipulator robot (myCobot 320).
- Participated in product concept brainstorming, development planning, and system behavior observation and analysis.

**Freelance Robotics and Embedded System Engineer** 2018-2019

- Built an embedded system for solar tracker to move the solar panel by given position that sent from base-station.
- Built a small-device for weighing system used in a gas company.
- Built embedded system for pull and press machine including monitoring of the pressure.

## SKILLS

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| **Programming** | C++ (primary), Python, C |
| **Robotic Systems** | ROS1/ROS2, MoveIt, ROS Navigation, SLAM, Path Tracking & Navigation, Sensor Fusion, Obstacle Avoidance, Learning-based Locomotion(IsaacLab) |
| **Real-time Control** | Embedded Systems, Serial Communication, Control Systems, PID/Fuzzy Controllers |
| **Computer Vision** | OpenCV, PyTorch, Object Detection and Instance Segmentation (YOLO and MaskRCNN), Visual Servoing, Camera Calibration |
| **Simulation** | MuJoCo, IsaacSim, PyBullet, Gazebo, V-REP |
| **Development Tools** | Git, CI/CD, Docker, Linux |
| **Design & Fabrication** | CAD (Fusion 360, SolidWorks), CAM, PCB Design (KiCad, EasyEDA, Eagle) |
| **Hardware** | ARM-based (STM32, NXP/Teensy), ESP32, ATMega, IMU, 2D and 3D Lidar, GNSS, Camera |
| **Robot Platforms** | UR3, Kachaka Pro, HR4C, myCobot 320, Kinova Jaco Dobot Nova 2, Seed-Noid-Mover |
| **Languages** | Indonesian (Native), English (Fluent), Japanese (Beginner) |

## PROJECTS

**Six-leg Robot Locomotion. January 2025 - December 2025**
Contribution:

- Developed high-level locomotion control using PPO reinforcement learning in MuJoCo simulation, with contributions to low-level control.

**Ornibibot, an agile flapping micro aerial vehicle (FMAVs). Oct 2019 - Present.**
This project develops an ornithopter that can be a helpful robot for future in the society. This project is a part of my academic path from master student to PhD student (current).

Contribution:

- Developed several robot's mechanical structure including design and manufacture.
- Developed robot's hardware utilizing several kinds of micro-controller (ARM based and ESP32) and single board computer (ARM based).
- Developed robot's low-level and high-level system and controller with or without ROS1/ROS2.

- Implemented PID controller with given Inertial Measurement data to perform attitude control using tail mechanism.
- Improved the performance serial communication by optimizing the packet data so it could work in above 500Hz.
- Conducted comprehensive system analysis by integrating force sensors and motion capture cameras to measure wing deformation and thrust generation.
- Improved system capabilities through mechanical optimization achieving 100g payload capacity, while implementing on board collision avoidance utilizing 8x8 ToF and 50Hz UDP communication with base station.
- Implemented outdoor localization by using GNSS and IMU sensor and tested VSLAM using ORB-SLAM3 or existing VIO algorithms.
- Enhancing collision avoidance performance by upgrading from 8x8 ToF sensor to PMD Flexx2 depth or point-cloud and validating the collision avoidance in MuJoCo simulation environment.
- Benchmarked potential usage of local 5G network for improved remote operation performance.

**Skills:** C, C++ (ROS1/ROS2), Python, CAD, CAM, EDA, MuJoCo, Computer Vision
**Source Code:** OrnibiBot — OrnibiBotMicro — ObstacleAvoidance — CollisionAvoidance — OrnibiBotESP

**Unmanned Water Surface Vehicle (January - March 2025)**
Developed autonomous water surface vehicle with advanced navigation and control systems, focusing on ROS2 migration and comprehensive system architecture design. Key Achievements:

- Migrated legacy ROS1 codebase to ROS2 for improved real-time performance and distributed system capabilities
- Debugged and optimized code and system for reliable autonomous operation (navigation and SLAM).

Skills: C++, ROS2

**Autonomous Service Robot with SLAM (May - July 2023)**
Led development of autonomous service robot for retail environment using manipulator, SLAM and computer vision.

Key Achievements:

- Led programming team for World Robot Summit competition (2nd runner-up).
- Integrated multiple subsystems: manipulator pick and place, LiDAR SLAM, Kinect vision, and navigation stack.
- Configured and tuned nav2 autonomous navigation stack with dynamic obstacle avoidance for retail environment.
- Integrated and improved object recognition pipeline using fine-tuned YOLO models for grasp pose estimation.

Skills: Python, ROS, MoveIt, PyTorch, SLAM, Computer Vision
Repository: WRS2023

**Autonomous Mobile Robot Navigation System - Bachelor Thesis (2017-2018)**
Developed advanced navigation system for omnidirectional robot soccer platform with autonomous path recovery capabilities after collision or obstacle interference.

Project Overview: This bachelor thesis project addressed the challenge of navigation recovery when robots collide with obstacles or other robots during autonomous operation. The system was validated through V-REP simulation environment by implementing bicycle path tracking algorithms.

Key Achievements:

- Implemented bicycle path tracking algorithm with fuzzy logic controller that dynamically adjusts look-ahead distance and robot velocity, significantly reducing overshoot during path recovery
- Developed real-time PID controllers for precise individual motor speed control and robot orientation
- Integrated wheel odometry and IMU data for accurate position estimation and localization

- Successfully demonstrated smooth path recovery after external force disturbances, with robot automatically returning to desired trajectory
- Designed and implemented ROS-based control architecture with 3 STM32 microcontrollers and 1 Arduino Mega microcontrollers(3 for motor control, 1 for odometry) each interfacing directly with ROS high-level planning nodes via serial communication.
- Validated system performance through simulation testing where external forces were applied to knock robot off-path, demonstrating autonomous recovery capabilities

Skills: C/C++, STM32, Embedded Systems, ROS, PID/Fuzzy Controllers, Qt, V-REP Simulation
Repository: FuzzyBicyclePathTracking

**Unmanned Fast Boat PENShip. Oct 2016 - Dec 2016.**
Built autonomous water surface vehicles with visual servoing navigation system, successfully demonstrating path-following capabilities in competitive environment.

- Implemented visual servoing control for centerline tracking using custom color-based detection in OpenCV, successfully competing with both catamaran and monohull platforms in USV Contest 2016.
- Led team coordination and system integration efforts

Skills: C/C++, Computer Vision (OpenCV), Embedded Systems

**ABU Robocon 2016**

Key Achievements:

- Developed embedded control systems for coordinated two-robot autonomous operation
- Implemented navigation algorithms using odometry feedback for precise robot positioning and guidance
- Integrated multiple feedback sensors (IMU, encoders, proximity sensors) for robust system operation
- Achieved significant performance improvements through iterative optimization: 36 seconds (regional), 23 seconds (national), and 18 seconds (international competition)
- Successfully demonstrated non-contact robot guidance, autonomous object manipulation, and climbing capabilities
- Led system integration, testing, and performance optimization achieving 70% improvement over previous competition results
- Won ABU Robocon Indonesia Contest 2016 and achieved 2nd Runner-up in international competition

Skills: C, STM32, Embedded Systems, Autonomous Navigation, Sensor Integration

## EDUCATION

**Doctor of Mechanical Systems Engineering**, Tokyo Metropolitan University — Mar 2026
**Master of Mechanical Systems Engineering**, Tokyo Metropolitan University — Oct 2019 - Sept 2021

- Relevant Courses: Robotic System Design, Intelligent Robot, Ubiquitous Robotics

**Bachelor of Computer Engineering**, Electronics Engineering Polytechnic Institute of Surabaya — 2014 - 2018

- Relevant Courses: Embedded System, Robot System & Control, Computer Vision, Intelligent System, Robotic and Automation, and Real-Time Operating System

## HONOR AND MEMBERSHIP

- Institute of Electrical and Electronics Engineers (IEEE) Society (current)
- IEEE Robotics and Automation Society Membership (current)
- The Japan Society of Mechanical Engineers (current)

- Second Runner-up Customer Interaction Task Category, World Robot Summit FCSC @IFAC 2023
- Runner Up Toilet Cleaning Robot Category, World Robot Summit FSCS Trial Competition 2019
- Second Runner Up of ABU Robocon 2016 2016
- Winner ABU Robocon Indonesia Contest 2016

## ACADEMIC EXPERIENCES

**Research Assistant** (Part-Time) Jan 2025 - Present
Tokyo Metropolitan University *Japan*

- Developing unmanned water surface vehicle with autonomous navigation, focusing on system integration, debugging, and ROS2 migration
- Developing 6-DoF lightweight manipulator including actuator using 3D-printed materials.
- Continuing flapping-wing aerial vehicle research and system optimization

**Tokyo Metropolitan University Arena Reframing (AR) Research Fellow** Apr 2023 - Sept 2024
Tokyo Metropolitan University *Japan*

- Conducted research on flapping-wing aerial vehicle development and performance optimization.

**Research Assistant** (Part-Time) Oct 2021 - Mar 2023
Tokyo Metropolitan University *Japan*

- Developed mechanical structure of servo-driven flapping-wing robot system and parameters observation.

**Student Intern** Jan 2017 - Feb 2017
BRIN-Aviation and Space Research Organization *Indonesia*

## PUBLICATIONS

### Journals

- Separate camber wing (SCW) device to enhance lift and payload on servo-driven flapping robot. (Co-author)
- Agile Maneuvering of Servo-Driven Flapping Robot: Novel Approach Using Center Flapping Angle and Offset Angle Variables. IEEE/ASME Transactions on Mechatronics. (Co-author)
- Performance Evaluation and Wing Deformation Analysis of Flapping-Wing Aerial Vehicles with Varying Flapping Parameters and Patterns, Journal of Robotics and Mechatronics, Vol.36, No.5, 2024/10.
- Senswing: A Force Sensing Wing for Intelligent Flapping-Wing Aerial Vehicles—Wing Design and Comprehensive Evaluation of Force Sensing Capabilities., Journal of Robotics and Mechatronics, Vol.36, No.5, 2024/10. (Co-author)
- Study Towards a Flapping Robot Maintaining Attitude During Gliding. International Journal on Advance Science, Engineering, Information, and Technology (IJASEIT) · Jan 22, 2023.

### Conferences

- Free-Flight Onboard Force Measurement for a Servo-Driven Ornithopter (AIM 2026). (Co-author)
- Servo-Driven Flapping Robot That Uses Its Tail for Self-Standing Takeoff. 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025). (Co-author)
- Servo-Driven Flapping-Wing Aerial Vehicle (FWAV) Payload Capacity and Navigation Performance. The 2025 IEEE/SICE International Symposium on System Integration (SII).

- Development of Flapping Robot with Self-Takeoff from The Ground Capability. IEEE International Conference on Robotics and Automation (ICRA) · Oct 18, 2021
- Bicycle Path Planning on Omnidirectional Mobile Robot Using Fuzzy Logic Controller. IEEE · Aug 7, 2018
- Implementation of PID Controller in Active Ball Handling System of Middle Size Robot Soccer. IEEE · Aug 7, 2018. (Co-author)
- Aksara Jawa Text Detection in Scene Images using Convolutional Neural Network. IEEE · Sep 27, 2017